

Idea

Riccardo Rebonato

EDHEC Business School – EDHEC Risk Institute

Outline

Our Goals

The Problem

The Strategy

Our Goals

- ▶ Portfolio managers use rules of thumbs to hedge with Treasuries their equity portfolios.
- ▶ For retail investors, typical recommendations are 60-40 allocation.
- ▶ We want to determine optimal *dynamic* (state dependent) allocations to equities, Nominal and Real Bonds so as to
 1. either optimize Sharpe Ratios;
 2. or minimize drawdowns;
 3. or maximize the Sortino Ratio;
 4. or ...

Our Goals

- ▶ We use an approach which is formally exactly equivalent to a one-layer neural network, but has the advantage of being fully interpretable.
- ▶ In particular the portfolio manager will be able to see the optimal weights that would have been determined over the historical period, and to see exactly which macrofinancial variables the dynamic weights depend on.
- ▶ So, the portfolio manager will not be faced with a black-box.

The Problem

- ▶ A portfolio manager wants to hedge her equity exposure by adding Treasuries to her portfolio.
- ▶ She has noticed that the automatic 'Greenspan put' does not necessarily apply in a world with rising inflation.
- ▶ She has also noticed that, when inflation expectation increase, the demand for TIPSs increases, and that it plummets when inflation expectations fall.
- ▶ She also knows that the price of TIPSs is affected by liquidity.

The Problem

- ▶ Therefore she wants to create a balanced portfolio of equities, Nominal and Real Bonds to maximize some performance criterion.
- ▶ In theory, this allocation should depend on the state of the world: so, for instance, she does not want to hope for Nominal Treasuries to come to the rescue when equities sag because of inflation fears.
- ▶ In practice, is a dynamic allocation strategy really better than a fixed-weight strategy out of sample? Or is she better off sticking with a simple 60-40 (or 30-70 or whatever) strategy?
- ▶ If dynamic weights are better, what determines the change in weights? What is the average weight value? How big are the fluctuations around this average?

The Problem

- ▶ She has \$1 to invest (long-only portfolio).
- ▶ Denoting S_t , P_t and T_t the prices of the S&P500, the price of the 10-year Nominal and the price of the 5-year TIPSs at time t , her goal is to allocate

$$\Pi(t) = \alpha \cdot S_t + \beta \cdot (P_t + T_t) \quad (1)$$

with

$$\begin{aligned} \alpha + \beta &= 1 \\ \beta_T &= (1 - \alpha) \cdot \gamma \\ \beta_P &= (1 - \alpha) \cdot (1 - \gamma) \end{aligned} \quad (2)$$

The Problem

- ▶ Putting all the pieces together, she wants an allocation

$$\Pi(t) = \alpha \cdot S_t + (1 - \alpha) \cdot \gamma \cdot T_t + (1 - \alpha) \cdot (1 - \gamma) \cdot P_T \quad (3)$$

with α and γ state-dependent coefficients:

$$\begin{aligned} \alpha &= \alpha(\omega) \\ \gamma &= \gamma(\omega) \end{aligned} \quad (4)$$

- ▶ The state of the world is characterized by a number of macrofinancial quantities (see below)

The Problem

- ▶ We take as characterization of the state of the world
 1. the inflation level at time t ;
 2. the change in inflation at time t ;
 3. a measure of volatility at time t ;
 4. the level of yields at time t .
- ▶ These variables are collectively denoted as $\{z\}$. So, $\omega = \{z\}$.
- ▶ Let the sigmoid (activation) function be denoted by $\Sigma(\cdot)$
- ▶ To begin with with assume that $0 \leq \alpha \leq 1$ and $0 \leq \gamma \leq 1$.
- ▶ It is easy to modify the procedure to allow, more realistically, for $\alpha_{min} \leq \alpha \leq \alpha_{max}$ and $\gamma_{min} \leq \gamma \leq \gamma_{max}$.

The Problem

- ▶ The state-dependent coefficients, $\alpha(z)$ and $\gamma(z)$, are the outputs of the sigmoid function:

$$\begin{aligned}\alpha(z) &= \Sigma(x(z)) \\ \gamma(z) &= \Sigma(y(z))\end{aligned}\tag{5}$$

and x and y (scalars) are linear functions of the macrofinancial variables (and, perhaps, interaction terms):

$$\begin{aligned}x(z) &= a_0 + \sum_{j=1}^4 a_j \cdot z_j \\ y(z) &= b_0 + \sum_{j=1}^4 b_j \cdot z_j\end{aligned}\tag{6}$$

The Problem

- Denote collectively

$$\eta = \{\{a\} \cup \{b\}\} \quad (7)$$

- Therefore we have

$$\begin{aligned} \alpha &= \Sigma(x(a)) \\ \gamma &= \Sigma(y(b)) \end{aligned} \quad (8)$$

- If we denote by χ our preferred performance measure, we then have

$$\chi = \chi(\alpha, \beta) = \chi(\Sigma(x(a)), \Sigma(y(b))) \quad (9)$$

- We want to maximize χ by searching over $\eta = \{\{a\} \cup \{b\}\}$.

The Strategy

- ▶ We do not have a lot of data. Therefore we estimate the optimal coefficients $\hat{\eta}$ using all the available data.
- ▶ Then we carry out a Gospodinov-like resampling, and create N equivalent universes, in which prices and macrofinancial variables are evolved consistently.
- ▶ We apply the coefficient $\hat{\eta}$ in each of the N universes, and, for each universe i , we compare the performance measure $\chi_i(\hat{\eta})$ with $\chi_i(\bar{\alpha}, \bar{\gamma})$, where $\bar{\alpha}$ and $\bar{\gamma}$ are **fixed (ie, non state dependent)** coefficients.

The Strategy

- ▶ We do this for all combinations of $\bar{\alpha}$ and $\bar{\gamma}$ (over a grid) to see whether the state-dependent allocation *with the coefficients for the dynamic weights determined out of sample* performs better than *any* fixed-weight allocation.
- ▶ In practice we will create a distribution of over- or under-performance (out of sample) for the dynamic-weight portfolio.